

Necessity of admissible angles for the periods $c + a$ and $c + b$ in three-move subtraction games

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Abstract

Let $1 \leq a < b < c$ and let g be the Sprague–Grundy sequence of the normal-play subtraction game with moves a, b, c . We prove that a period $c + a$ or $c + b$ of the zero/nonzero outcome sequence from zero forces the corresponding admissible-angle condition of Manabe’s Definition 19. Failure of the first condition gives $g(x) = 0 < g(x + c + a)$ for some $0 \leq x < b - a$; failure of the second gives $g(x) = 0 < g(x + c + b)$ for some $0 \leq x < a$. The proof evaluates the actual zero positions immediately above c and establishes a difference-set identity modulo $a + b$. Consequently, both least–Sprague–Grundy-period implications in Conjecture 55 of arXiv:2609.05358v1 hold. The argument does not require coprimality, nonadditivity, or leastness of the nominated period.

1 Statement and context

In a normal-play subtraction game, a position is a heap of $n \in \mathbb{Z}_{\geq 0}$ tokens, and a move subtracts an allowed positive integer without making the heap negative. For $S = \{a, b, c\}$ its Sprague–Grundy sequence is

$$g(n) = \text{mex}\{g(n - s) : s \in S, s \leq n\}, \quad (1)$$

where mex denotes the least nonnegative integer absent from a set; in particular, $\text{mex } \emptyset = 0$. Thus $g(n) = 0$ precisely when every legal option has nonzero value. Define the outcome sequence by $o(n) = 0$ when $g(n) = 0$ and $o(n) = 1$ otherwise. A period P from zero means equality at n and $n + P$ for every $n \geq 0$. A period of g is therefore a period of o .

Put

$$L = a + b, \quad \delta = b - a = \eta a + \varepsilon, \quad 0 \leq \varepsilon < a, \quad \rho = c \bmod L.$$

All intervals below contain integer points and are half-open. Residue sets are represented in $[0, L)$. Their translations and Minkowski differences are taken in $\mathbb{Z}/L\mathbb{Z}$; specifically, $X - Y = \{x - y \bmod L : x \in X, y \in Y\}$. Ordinary set removal is written $X \setminus Y$. Define

$$Z = \{r \in [0, b) : \lfloor r/a \rfloor \text{ is even}\}, \quad D = [a, b) \setminus Z, \quad (2)$$

$$\Lambda = \begin{cases} [(\eta + 1)a, (\eta + 1)a + \varepsilon), & \eta \text{ even}, \\ [\eta a + \varepsilon, (\eta + 1)a), & \eta \text{ odd}. \end{cases} \quad (3)$$

The set Z is exactly the zero-residue set of the two-move base game $\{a, b\}$, as verified below. In this notation the two relevant conditions in Manabe’s Definition 19 [1] are

$$\rho \in \Theta_{c+a} \iff Z \cap [0, \delta) \cap (Z + \delta - \rho) = \emptyset, \quad (\text{T1})$$

$$\rho \in \Theta_{c+b} \iff (\rho + \Lambda) \cap Z = \emptyset \quad \text{and} \quad (\rho + D) \cap \Lambda = \emptyset. \quad (\text{T2})$$

These formulas specify the notation Θ completely for the two periods studied here.

Theorem 1. *For every integer triple $1 \leq a < b < c$, the following assertions hold.*

1. *If (T1) fails, there is an integer $0 \leq x < \delta$ such that $g(x) = 0$ and $g(x + c + a) > 0$.*
2. *If (T2) fails, there is an integer $0 \leq x < a$ such that $g(x) = 0$ and $g(x + c + b) > 0$.*

Consequently, if $c + a$ is an outcome period from zero then $\rho \in \Theta_{c+a}$, and if $c + b$ is such a period then $\rho \in \Theta_{c+b}$.

Corollary 2 (Manabe’s Conjecture 55). *Let $2 \leq a < b < c$, $c \neq a + b$, and $\gcd(a, b, c) = 1$. Suppose that g is purely periodic from zero with least period p . If $p = c + a$, then $\rho \in \Theta_{c+a}$; if $p = c + b$, then $\rho \in \Theta_{c+b}$.*

Manabe [1, Theorem 23] proves a sufficient admissibility criterion for outcome periodicity and, in Theorem 32, determines the least nim period from the least admissible candidate under the original hypotheses. Conjecture 55 asks for the two necessity implications in Corollary 2. The displacement $c - \delta$ used for the $c + a$ branch already occurs in the anti-collision argument of that source’s Theorem 23. For the $c + b$ branch, the additional work here is to compute the actual zero set in $[c, c + b)$ without assuming a prescribed periodic pattern, and to prove the circular difference identity in Lemma 5.

Miklós and Post [2, Lemma 2.13 and Corollary 2.12.1] give translated-zero and finite-window criteria for outcome periodicity. Their Theorems 3.3–3.4 describe the two-move outcome word. These criteria and the base word are prior infrastructure. Their Conjecture 3 uses a separate division-algorithm predicate; equivalence with (T2) is not established here. At a broader methodological level, exact period proofs with finite computational corroboration also occur in Zeng’s work on reversible cellular automata [3]; that work concerns a different dynamical system.

The comparison above is with the cited versions inspected on 9 September 2026 and makes no comprehensive priority claim. The result settles Conjecture 55 as stated in [1]; it does not settle Conjecture 54, the remaining small- c necessity question for $a + b$, or the full classification of purely periodic three-move games.

2 The base word and the first band above c

Lemma 3. *The zero positions of the two-move game $\{a, b\}$ are precisely the nonnegative integers whose residues modulo L lie in Z . Moreover,*

$$\Lambda = D \cap [\delta, b), \tag{4}$$

$$\{r \in \mathbb{Z}/L\mathbb{Z} : r + a \notin Z\} = Z \cup \Lambda. \tag{5}$$

Proof. Below b only the move a can be available, so the outcome word alternates between zero and nonzero blocks of length a . Every position in $[b, L)$ has a b -move to $[0, a)$ and is nonzero. This proves the claimed pattern on its first block of length L .

For $r \in Z$, both $r - a$ and $r + a$, reduced modulo L , are outside Z : the predecessor lies in the preceding odd block, or in $[b, L)$ if $r < a$; the successor lies in the next odd block, or in $[b, L)$. Since $b \equiv -a \pmod{L}$, neither move joins two proposed zeros. A residue outside Z and below b has an a -predecessor in an even block; a residue in $[b, L)$ has a b -predecessor in $[0, a)$. For positions at least L both moves are legal. Induction using the zero-position recurrence now proves the asserted periodic base word.

Intersecting the odd blocks with the final length- a window $[\delta, b) = [\eta a + \varepsilon, (\eta + 1)a + \varepsilon)$ gives exactly (3), proving (4). For (5), a residue in Z has its a -successor outside Z . A residue $r \in [b, L)$

has $r + a \equiv r - b \in [0, a) \subset Z$. Finally, if $r \in D$, then $r + a$ lies in the next even block precisely when $r + a < b$; otherwise it lies in $[b, L)$. The latter case is exactly $r \geq \delta$. These cases partition all residues and prove the identity. $\square$

Lemma 4. *For every integer $0 \leq t < b$,*

$$g(c+t) = 0 \iff t \in D \text{ and } \rho + t \in Z \cup \Lambda \pmod{L}. \quad (6)$$

Proof. Below c , the three-move game agrees with the base game. If $t \notin D$, then $0 \leq t < b$ implies $t \in Z$, so the c -move from $c+t$ reaches a zero and $g(c+t) > 0$.

If $t \in D$, then $t \geq a$, $t - a \in Z$, and $g(t) > 0$. The a -option $c+t-a$ itself has a legal c -move to $t-a$, so $g(c+t-a) > 0$ as well. The remaining option $c+t-b$ belongs to $[c-b, c)$, which is nonnegative and below c . Thus $g(c+t) = 0$ exactly when that option is a base nonzero. Its residue is $\rho + t - b \equiv \rho + t + a \pmod{L}$, so (5) gives (6). $\square$

3 A circular difference identity

Lemma 5. *With the sets in (2)–(3),*

$$(Z - \Lambda) \cup (\Lambda - D) = (Z \cup \Lambda) - \Lambda \quad \text{in } \mathbb{Z}/L\mathbb{Z}. \quad (7)$$

Proof. Since $\Lambda \subseteq D$, the right-hand side is contained in the left-hand side. For the converse it is enough to prove

$$\Lambda - D \subseteq (Z - \Lambda) \cup (\Lambda - \Lambda). \quad (8)$$

For nonempty integer intervals, their ordinary difference is

$$[u, v] - [r, s] = [u - s + 1, v - r]. \quad (9)$$

Indeed the endpoints are the least and greatest attainable integer differences, and every intervening integer occurs. The interval on the right is subsequently reduced modulo L when used on the circle.

Even case: $\eta = 2h$. Here

$$\begin{aligned} L &= (2h+2)a + \varepsilon, & Z &= \bigcup_{i=0}^h [2ia, (2i+1)a), \\ D &= \bigcup_{j=0}^{h-1} [(2j+1)a, (2j+2)a) \cup \Lambda, & \Lambda &= [(2h+1)a, (2h+1)a + \varepsilon). \end{aligned}$$

If $\varepsilon = 0$, then Λ is empty and (8) holds. Otherwise, for $0 \leq j < h$ set $i = h - j - 1$. By (9),

$$\begin{aligned} \Lambda - [(2j+1)a, (2j+2)a) &= [(2h-2j-1)a+1, (2h-2j)a+\varepsilon) \\ &\equiv [2ia, (2i+1)a) - \Lambda \pmod{L}. \end{aligned}$$

The second line follows by adding L to the ordinary difference on that line. These contributions lie in $Z - \Lambda$; the remaining contribution from D is $\Lambda - \Lambda$. If $h = 0$, the union over j is empty and the same argument applies.

Odd case: $\eta = 2h + 1$. Now

$$\begin{aligned} L &= (2h + 3)a + \varepsilon, \\ Z &= \bigcup_{i=0}^h [2ia, (2i + 1)a) \cup Q, & Q &= [(2h + 2)a, (2h + 2)a + \varepsilon), \\ D &= \bigcup_{j=0}^h [(2j + 1)a, (2j + 2)a), & \Lambda &= [(2h + 1)a + \varepsilon, (2h + 2)a). \end{aligned}$$

Here Λ is nonempty because $\varepsilon < a$. With $d = h - j$, the difference of Λ and the j th block of D is

$$I_d = [(2d - 1)a + \varepsilon + 1, (2d + 1)a), \quad 0 \leq d \leq h.$$

For $d \geq 1$, take $i = d - 1$. Formula (9), followed by adding L , gives $I_d \equiv [2ia, (2i + 1)a) - \Lambda \pmod{L}$, so $I_d \subseteq Z - \Lambda$ on the circle. At $d = 0$,

$$I_0 = [-a + \varepsilon + 1, a), \quad \Lambda - \Lambda = [-a + \varepsilon + 1, a - \varepsilon).$$

If $\varepsilon = 0$, these intervals coincide. If $\varepsilon > 0$, then Q is nonempty and (9) gives $Q - \Lambda = [1, a)$. Since $1 \leq a - \varepsilon$, this contains the remaining interval $[a - \varepsilon, a)$. Thus (8) also holds in the odd case, proving (7). No difference formula has been applied to an empty interval. $\square$

4 Witnesses and completion of the proof

Proof of Theorem 1. If (T1) fails, choose $x \in Z \cap [0, \delta) \cap (Z + \delta - \rho)$ and set $y = c + x - \delta$. Since $c > b$,

$$0 < c - \delta \leq y < c, \quad 0 \leq x < \delta < c.$$

The residue of y is $\rho + x - \delta \in Z$. Agreement with the base game below c gives $g(x) = g(y) = 0$. The legal b -move

$$x + c + a \mapsto x + c + a - b = y$$

therefore shows that $g(x + c + a) > 0$.

Failure of (T2) is equivalent to

$$\rho \in (Z - \Lambda) \cup (\Lambda - D).$$

By Lemma 5, choose $t \in \Lambda$ with $\rho + t \in Z \cup \Lambda \pmod{L}$. Lemma 4 gives $g(c + t) = 0$. Equation (4) yields $\delta \leq t < b$, so $x = t - \delta$ satisfies $0 \leq x < a$ and $g(x) = 0$. The legal a -move

$$x + c + b \mapsto x + c + b - a = c + t$$

shows that $g(x + c + b) > 0$. In either branch the mismatch excludes the nominated period of the outcome sequence from zero. $\square$

Proof of Corollary 2. A period of g preserves its zero positions, so Theorem 1 applies to each of the two possible values of p . $\square$

Example. For $(a, b) = (3, 8)$ one has $L = 11$, $\delta = 5$, $Z = \{0, 1, 2, 6, 7\}$, $D = \{3, 4, 5\}$, and $\Lambda = \{5\}$. For every $c > 8$ with $c \neq 11$, the following entries give witness indices x from Theorem 1:

ρ	0	1	2	3	4	5	6	7	8	9	10
Shift $c + 3$	0	0	–	2	1	0	0	0	–	2	1
Shift $c + 8$	0	0	0	–	–	–	0	0	0	–	–

An entry means $g(x) = 0$ and the shifted value is positive; a dash means the angle is admissible in that column and asserts no least period. In the first row the 8-move reaches the zero $c + x - 5 < c$. In the second row $t = 5$ gives the zero $c + 5$, reached from $c + 8$ by a 3-move. Thus these formulas hold for every winding of c modulo 11, rather than only a finite list of c .

Computational corroboration and assistance. The proof is deductive and uses no computational bound. Exact Python integer computations supplied additional checks: a separately written verifier checked the base and difference identities for all 10,208 pairs $1 \leq a \leq 64$, $a < b \leq 192$, and the seam and witness formulas for all 28,245 triples $1 \leq a \leq 14$, $a < b \leq 52$, $b < c \leq 2(a + b) + 1$, without exclusions. It also replayed 264 serialized cases containing 309 witnesses and 12 complete mex words. No assertion failed. These finite checks corroborate the proof; they do not establish its universal quantifiers. The accompanying verification dossier retains the programs, inputs, and exact outputs. OpenAI Codex assisted with proof development, exact verification, manuscript drafting, and a separate mathematical review pass. That review was AI-assisted and is not external peer review. No formal proof-assistant verification is claimed.

References

- [1] Hikaru Manabe, *Purely Periodic Three-move Subtraction Games*, arXiv:2609.05358v1 (2026). <https://arxiv.org/abs/2609.05358v1>.
- [2] István Miklós and Logan Post, *Superpolynomial period lengths of the winning positions in the subtraction game*, arXiv:2312.02426v1 (2023). <https://arxiv.org/abs/2312.02426v1>.
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